

KARUKURI HITESH VARMA

hiteshvarmakarukuri@gmail.com — linkedin.com/in/hitesh-varma-karukuri — github.com/hiteshvarmak

PROFESSIONAL SUMMARY

Computer Science and Data Science graduate with hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and software development. Experienced in Python, C++, Java, SQL, web technologies, data analysis, and visualization, with practical exposure through an AI internship and projects involving real-time surveillance, data aggregation, and web-based automation. Strong foundation in Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, and Operating Systems.

TECHNICAL SKILLS

Programming Languages	Python, C++, Java
AI & Machine Learning	Machine Learning, Deep Learning, Computer Vision, MobileNetV3, ConvLSTM
Data Science	Pandas, R, SQL, Data Analysis
Software Development	Node.js, Object-Oriented Programming, Application Development
Web Technologies	HTML, CSS, JavaScript
Databases	MySQL, Database Management
Data Visualization	Power BI, Tableau, Excel
Version Control & Tools	Git , GitHub
Core CS	Data Structures and Algorithms, Computer Networks, Operating Systems, Database Management Systems

EXPERIENCE

Swecha – Meta Summer of AI Internship <i>AI Intern, Remote</i>	May 2025 – June 2025
- Applied Artificial Intelligence and Machine Learning techniques to develop data-driven solutions for real-world challenges as part of a 4-week AI internship focused on the Global South. - Performed data preprocessing, exploratory data analysis, feature engineering, model development, and evaluation across end-to-end machine learning workflows. - Gained hands-on experience in machine learning workflows, data preprocessing, model development, and AI-based problem solving.	

PROJECTS

MobileNet-ConvLSTM Based Intelligent Border Surveillance System [GitHub] <i>Python, Deep Learning, Computer Vision, Node.js, IoT</i>	September 2025
- Developed a low-power AI-based spatio-temporal surveillance system for real-time border security using edge devices. - Built a hybrid MobileNetV3-ConvLSTM model for spatio-temporal analysis to reduce false alarms. - Architected a two-stage pipeline: edge-level intrusion detection triggers GPU-accelerated secondary verification (super-resolution + ResNet-based bounding-box detection) via a Node.js server, cutting false-positive escalations by 5.2%. - Filed a patent for the system (Feb 2026) and authored a manuscript currently under peer review, addressing reviewer revisions across model justification and evaluation methodology.	
Hostel Daily Pass System – Digital Token Generator [GitHub] <i>HTML, CSS, JavaScript, Local Storage</i>	February 2025
- Designed and developed a web-based digital hostel pass system that generates unique, time-bound access tokens for registered hostellers. - Implemented client-side token validation and QR-based verification to strengthen hostel access control and improve resource utilization. - Achieved a reported 100% reduction in non-hosteller access to hostel mess facilities after implementation.	
Healthcare Patient Data Aggregation System <i>Python, MySQL, Pandas</i>	August 2024
- Built a data pipeline to consolidate patient records from multiple hospital branches into a unified database. - Designed database schemas and handled missing and duplicate records to improve data consistency. - Used SQL-based analysis to generate insights from the consolidated healthcare dataset.	

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology

B.Tech. in Computer Science and Engineering – Data Science

2022–2026

CGPA: 8.99/10

Pinegrove Junior College

Senior Secondary

2020–2022

Percentage: 97.4%

Paramita Heritage School

Secondary School

2019

CGPA: 10.0/10

ACHIEVEMENTS

• **Patent – MobileNet-ConvLSTM-Based Intelligent Border Surveillance System**

February 2026

• **Finalist – SDG Hackathon – IIIT Hyderabad**

September 2024

CERTIFICATIONS

• **Agentic AI Certified Foundations Associate – Oracle**

August 2026

• **Advanced SQL – SDG Hackathon, IIIT Hyderabad**

February 2025

• **Power BI Essentials – LinkedIn Learning**

February 2025

• **Principles of Generative AI – Foundational Course**

July–October 2024

LEADERSHIP & ACTIVITIES

Design Lead, VJ Data Questers

March 2024 – May 2026

• *Led visual design strategy for the technical club, translating event objectives into consistent and high-impact visual communication across club initiatives.*

• *Coordinated design activities and contributed to maintaining a consistent visual identity across events, promotional campaigns, and organizational initiatives.*

Volunteer, VNR Student Force

April 2023 – May 2026

• *Contributed to community outreach initiatives through the collection, organization, and on-ground distribution of resources for social service activities.*

• *Collaborated with volunteers to coordinate activities and support the execution of community-focused initiatives.*

Volunteer, Vignana Jyothi Sahithivanam

August 2025 – May 2026

• *Supported the planning and execution of literary events, contributing to event coordination, on-ground logistics, and attendee engagement.*

• *Assisted with event activities and coordination during literary fests and organizational initiatives.*